

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12405473
B2
September 02, 2025
Bhakta; Vikrant et al.

Optical systems with reflective prism input couplers

Abstract

An electronic device may include a display module that generates light and an optical system that redirects the light towards an eye box. The system may include an input coupler on a waveguide and a lens that directs the light towards the input coupler. The input coupler may include a prism having a reflective surface that reflects the light into the waveguide. The reflective surface may be curved to provide the light with an optical power. The prism may be configured to expand a field of view of the light. A birefringent beam displacer may expand the effective pupil size of the light. The lens may include lens elements that converge the light at a location between the lens elements and the waveguide. A switchable panel may be placed at the location and toggled between first and second orientations to increase the effective resolution of the light.

Inventors: Bhakta; Vikrant (Santa Clara, CA), Peng; Guolin (Sunnyvale, CA), Choi; Hyungryul (San Jose, CA), Keilbach; Kevin A. (Boulder, CO), DeLapp; Scott M. (San Diego, CA)

Applicant: Apple Inc. (Cupertino, CA)

Family ID: 72659366

Appl. No.: 18/641244

Filed: April 19, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240272438 A1	Aug. 15, 2024

Related U.S. Application Data

continuation parent-doc US 17477321 20210916 US 11994681 child-doc US 18641244
continuation parent-doc WO PCT/US2020/050709 20200914 PENDING child-doc US 17477321
us-provisional-application US 62902645 20190919

Publication Classification

Int. Cl.: G02B6/00 (20060101); F21V8/00 (20060101); G02B27/01 (20060101)

U.S. Cl.:

CPC G02B27/0172 (20130101); G02B6/0016 (20130101); G02B6/003 (20130101);
G02B2027/0123 (20130101); G02B2027/0147 (20130101); G02B2027/0163 (20130101)

Field of Classification Search

CPC: G02B (27/0172); G02B (6/0018); G02B (6/0023)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4405941	12/1982	Mecklenborg	N/A	N/A
6646809	12/2002	Ishino et al.	N/A	N/A
7418170	12/2007	Mukawa et al.	N/A	N/A
7502168	12/2008	Akutsu et al.	N/A	N/A
7764413	12/2009	Levola	N/A	N/A
8160411	12/2011	Levola et al.	N/A	N/A
8233204	12/2011	Robbins et al.	N/A	N/A
8446675	12/2012	Wang et al.	N/A	N/A
8665178	12/2013	Wang	N/A	N/A
8903207	12/2013	Brown et al.	N/A	N/A
9075184	12/2014	Popovich et al.	N/A	N/A
9341846	12/2015	Popovich et al.	N/A	N/A
9456744	12/2015	Popovich et al.	N/A	N/A
11119261	12/2020	Ayres et al.	N/A	N/A
11947110	12/2023	Pfeiffer et al.	N/A	N/A
2002/0021498	12/2001	Ohtaka et al.	N/A	N/A
2004/0227923	12/2003	Flagello et al.	N/A	N/A
2006/0268421	12/2005	Shimizu et al.	N/A	N/A
2013/0250430	12/2012	Robbins et al.	N/A	N/A
2014/0140653	12/2013	Brown et al.	N/A	N/A
2014/0140654	12/2013	Brown et al.	N/A	N/A
2014/0240843	12/2013	Kollin	N/A	N/A
2015/0219898	12/2014	Ko	359/630	G02B 27/0172
2015/0260992	12/2014	Luttmann et al.	N/A	N/A
2016/0041387	12/2015	Valera et al.	N/A	N/A
2016/0154150	12/2015	Simmonds et al.	N/A	N/A
2017/0031160	12/2016	Popovich et al.	N/A	N/A
2017/0276940	12/2016	Popovich et al.	N/A	N/A
2019/0094981	12/2018	Bradski et al.	N/A	N/A
2019/0113829	12/2018	Waldern et al.	N/A	N/A
2019/0391408	12/2018	Mansharof	N/A	N/A
2020/0103660	12/2019	Mukawa	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1539095	12/2003	CN	N/A
102576153	12/2011	CN	N/A
106716223	12/2016	CN	N/A
107272198	12/2016	CN	N/A
108693641	12/2017	CN	N/A
108700697	12/2017	CN	N/A
108873332	12/2017	CN	N/A
H07270791	12/1994	JP	N/A
H09230320	12/1996	JP	N/A
H11039706	12/1998	JP	N/A
2005062297	12/2004	JP	N/A
2005189677	12/2004	JP	N/A
2006126238	12/2005	JP	N/A
2007011279	12/2006	JP	N/A
2007052086	12/2006	JP	N/A
2017520013	12/2016	JP	N/A
2014155096	12/2013	WO	N/A
2015162611	12/2014	WO	N/A
2017077934	12/2016	WO	N/A
2017176389	12/2016	WO	N/A
2017176393	12/2016	WO	N/A

Primary Examiner: Park; Sanghyuk

Attorney, Agent or Firm: Treyz Law Group, P.C.

Background/Summary

(1) This application is a continuation of U.S. non-provisional patent application No. 17/477,321, filed Sep. 16, 2021, which is a continuation of international patent application No. PCT/US2020/050709, filed Sep. 14, 2020, which claims the benefit of U.S. provisional patent application No. 62/902,645, filed Sep. 19, 2019, which are hereby incorporated by reference herein in their entireties.

BACKGROUND

- (1) This relates generally to optical systems and, more particularly, to optical systems for displays.
- (2) Electronic devices may include displays that present images to a user's eyes. For example, devices such as virtual reality and augmented reality headsets may include displays with optical elements that allow users to view the displays.
- (3) It can be challenging to design devices such as these. If care is not taken, the components used in displaying content may be unsightly and bulky and may not exhibit desired levels of optical performance.

SUMMARY

- (4) An electronic device such as a head-mounted device may have one or more near-eye displays that produce images for a user. The head-mounted device may be a pair of virtual reality glasses or may be an augmented reality headset that allows a viewer to view both computer-generated images

and real-world objects in the viewer's surrounding environment.

(5) The near-eye display may include a display module that generates light and an optical system that redirects the light from the display module towards an eye box. The optical system may include an input coupler and an output coupler formed on a waveguide. The input coupler may redirect light from the display module so that the light propagates in the waveguide towards the output coupler. The output coupler may couple the light out of the waveguide and towards the eye box. The output coupler may include diffractive grating structures such as volume holograms in the waveguide.

(6) The input coupler may be a reflective input coupling prism mounted to the waveguide. A lens may direct the light from the display module to the input coupler through the waveguide. The prism may receive the light from the lens through the waveguide and may have a reflective surface that reflects the light into the waveguide. The reflective surface may be curved to provide the reflected light with an optical power. This may allow the lens to have fewer lens elements than in scenarios where the prism does not impart optical power on the light.

(7) The prism may be configured to expand a field of view of the light. For example, a switchable reflective layer may be provided on the reflective surface for expanding a field of view of the light. If desired, the prism may be a split prism that includes a first wedge having a first reflective surface and a second wedge mounted to the first reflective surface. The first reflective surface may reflect the image light in a first direction whereas the second reflective surface reflects the image light in a second direction. A polarization-sensitive film may be provided on the first reflective surface and may reflect a first polarization of light while transmitting a second polarization of light. A switchable polarizer may be used to sequentially provide the prism with light of the first and second polarizations. In another suitable arrangement, the display panel may concurrently provide the prism with light of the first and second polarizations. If desired, a switchable reflective layer may be provided on the first reflective surface and may selectively reflect or pass the light from the display module. If desired, a birefringent beam displacer may be interposed on the optical path of the display for expanding the effective pupil size of the light.

(8) The lens may include a first set of lens elements that converge the light from the display module at a location between the first set of lens elements and the waveguide. A transparent panel that is switchable between first and second orientations may be placed at the location. The transparent panel may be toggled between the first and second orientations to increase the effective resolution of the light. If desired, a second set of lens elements may be interposed between the location and the waveguide.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of an illustrative system having a display in accordance with some embodiments.

(2) FIG. 2 is a top view of an illustrative optical system for a display having a waveguide with an input coupler in accordance with some embodiments.

(3) FIG. 3 is a top view of an illustrative input coupler formed from a reflective prism in accordance with some embodiments.

(4) FIG. 4 is a top view of an illustrative input coupler formed from a reflective prism having a curved reflective surface in accordance with some embodiments.

(5) FIG. 5 is a top view of an illustrative input coupler formed from a reflective prism having a switchable reflective component for expanding the field of view for a display in accordance with some embodiments.

(6) FIG. 6 is a top view of an illustrative input coupler formed from a split reflective prism having

a polarization-sensitive reflective film for expanding a field of view in accordance with some embodiments.

(7) FIG. 7 is a top view of an illustrative input coupler formed from a split reflective prism having a switchable reflective component for expanding a field of view in accordance with some embodiments.

(8) FIG. 8 is a top view of an illustrative display system in which light of different polarizations is concurrently provided to a split reflective prism having a polarization-sensitive reflective film for expanding a field of view in accordance with some embodiments.

(9) FIG. 9 is a perspective view of an illustrative birefringent beam displacer that may be provided on a waveguide for expanding a pupil of light within the waveguide in accordance with some embodiments.

(10) FIG. 10 is a top view showing how an illustrative birefringent beam displacer of the type shown in FIG. 9 may pass light to an input coupler having a split reflective prism in accordance with some embodiments.

(11) FIG. 11 is a top view of an illustrative split multi-element lens that may provide image light to an input coupler of the types shown in FIGS. 2-10 in accordance with some embodiments.

(12) FIG. 12 is a top view showing how an illustrative switchable light displacing element may be mounted within a split multi-element lens of the type shown in FIG. 11 in accordance with some embodiments.

(13) FIG. 13 is a top view showing how a switchable light displacing element of the type shown in FIG. 12 may be adjusted between first and second states for displacing image light in accordance with some embodiments.

(14) FIG. 14 is a front view of pixels of image light that illustrates how an illustrative switchable light displacing element of the type shown in FIGS. 12 and 13 may increase the effective resolution of the image light in accordance with some embodiments.

(15) FIG. 15 is a top view showing how an input coupler formed from a reflective prism with a curved reflective surface may form part of a split multi-element lens in accordance with some embodiments.

(16) FIG. 16 is a top view showing how an input coupler formed from a reflective prism may receive image light through a dispersion compensation wedge and/or may include an optical absorber in accordance with some embodiments.

DETAILED DESCRIPTION

(17) An illustrative system having a device with one or more near-eye display systems is shown in FIG. 1. System 10 may be a head-mounted device having one or more displays such as near-eye displays 14 mounted within support structure (housing) 20. Support structure 20 may have the shape of a pair of eyeglasses (e.g., supporting frames), may form a housing having a helmet shape, or may have other configurations to help in mounting and securing the components of near-eye displays 14 on the head or near the eye of a user. Near-eye displays 14 may include one or more display modules such as display modules 14A and one or more optical systems such as optical systems 14B. Display modules 14A may be mounted in a support structure such as support structure 20. Each display module 14A may emit light 22 (image light) that is redirected towards a user's eyes at eye box 24 using an associated one of optical systems 14B.

(18) The operation of system 10 may be controlled using control circuitry 16. Control circuitry 16 may include storage and processing circuitry for controlling the operation of system 10. Circuitry 16 may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry 16 may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code (instructions) may be stored on

storage in circuitry **16** and run on processing circuitry in circuitry **16** to implement operations for system **10** (e.g., data gathering operations, operations involving the adjustment of components using control signals, image rendering operations to produce image content to be displayed for a user, etc.).

(19) System **10** may include input-output circuitry such as input-output devices **12**. Input-output devices **12** may be used to allow data to be received by system **10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, or other electrical equipment) and to allow a user to provide head-mounted device **10** with user input. Input-output devices **12** may also be used to gather information on the environment in which system **10** (e.g., head-mounted device **10**) is operating. Output components in devices **12** may allow system **10** to provide a user with output and may be used to communicate with external electrical equipment. Input-output devices **12** may include sensors and other components **18** (e.g., image sensors for gathering images of real-world object that are digitally merged with virtual objects on a display in system **10**, accelerometers, depth sensors, light sensors, haptic output devices, speakers, batteries, wireless communications circuits for communicating between system **10** and external electronic equipment, etc.).

(20) Display modules **14A** may include reflective displays (e.g., liquid crystal on silicon (LCOS) displays, digital-micromirror device (DMD) displays, or other spatial light modulators), emissive displays (e.g., micro-light-emitting diode (uLED) displays, organic light-emitting diode (OLED) displays, laser-based displays, etc.), or displays of other types. Light sources in display modules **14A** may include uLEDs, OLEDs, LEDs, lasers, combinations of these, or any other desired light-emitting components.

(21) Optical systems **14B** may form lenses that allow a viewer (see, e.g., a viewer's eyes at eye box **24**) to view images on display(s) **14**. There may be two optical systems **14B** (e.g., for forming left and right lenses) associated with respective left and right eyes of the user. A single display **14** may produce images for both eyes or a pair of displays **14** may be used to display images. In configurations with multiple displays (e.g., left and right eye displays), the focal length and positions of the lenses formed by components in optical system **14B** may be selected so that any gap present between the displays will not be visible to a user (e.g., so that the images of the left and right displays overlap or merge seamlessly).

(22) If desired, optical system **14B** may contain components (e.g., an optical combiner, etc.) to allow real-world image light from real-world images or objects **25** to be combined optically with virtual (computer-generated) images such as virtual images in image light **22**. In this type of system, which is sometimes referred to as an augmented reality system, a user of system **10** may view both real-world content and computer-generated content that is overlaid on top of the real-world content. Camera-based augmented reality systems may also be used in device **10** (e.g., in an arrangement which a camera captures real-world images of object **25** and this content is digitally merged with virtual content at optical system **14B**).

(23) System **10** may, if desired, include wireless circuitry and/or other circuitry to support communications with a computer or other external equipment (e.g., a computer that supplies display **14** with image content). During operation, control circuitry **16** may supply image content to display **14**. The content may be remotely received (e.g., from a computer or other content source coupled to system **10**) and/or may be generated by control circuitry **16** (e.g., text, other computer-generated content, etc.). The content that is supplied to display **14** by control circuitry **16** may be viewed by a viewer at eye box **24**.

(24) FIG. **2** is a top view of an illustrative display **14** that may be used in system **10** of FIG. **1**. As shown in FIG. **2**, near-eye display **14** may include one or more display modules such as display module **14A** and an optical system such as optical system **14B**. Optical system **14B** may include optical elements such as one or more waveguides **26**. Waveguide **26** may include one or more stacked substrates (e.g., stacked planar and/or curved layers sometimes referred to herein as

waveguide substrates) of optically transparent material such as plastic, polymer, glass, etc.

(25) If desired, waveguide **26** may also include one or more layers of holographic recording media (sometimes referred to herein as holographic media, grating media, or diffraction grating media) on which one or more diffractive gratings are recorded (e.g., holographic phase gratings, sometimes referred to herein as holograms). A holographic recording may be stored as an optical interference pattern (e.g., alternating regions of different indices of refraction) within a photosensitive optical material such as the holographic media. The optical interference pattern may create a holographic phase grating that, when illuminated with a given light source, diffracts light to create a three-dimensional reconstruction of the holographic recording. The holographic phase grating may be a non-switchable diffractive grating that is encoded with a permanent interference pattern or may be a switchable diffractive grating in which the diffracted light can be modulated by controlling an electric field applied to the holographic recording medium. Multiple holographic phase gratings (holograms) may be recorded within (e.g., superimposed within) the same volume of holographic medium if desired. The holographic phase gratings may be, for example, volume holograms or thin-film holograms in the grating medium. The grating media may include photopolymers, gelatin such as dichromated gelatin, silver halides, holographic polymer dispersed liquid crystal, or other suitable holographic media.

(26) Diffractive gratings on waveguide **26** may include holographic phase gratings such as volume holograms or thin-film holograms, meta-gratings, or any other desired diffractive grating structures. The diffractive gratings on waveguide **26** may also include surface relief gratings formed on one or more surfaces of the substrates in waveguides **26**, gratings formed from patterns of metal structures, etc. The diffractive gratings may, for example, include multiple multiplexed gratings (e.g., holograms) that at least partially overlap within the same volume of grating medium (e.g., for diffracting different colors of light and/or light from a range of different input angles at one or more corresponding output angles).

(27) Optical system **14B** may include collimating optics such as collimating lens **34**. Collimating lens **34** may include one or more lens elements that help direct image light **22** towards waveguide **26**. If desired, display module **14A** may be mounted within support structure **20** of FIG. **1** while optical system **14B** may be mounted between portions of support structure **20** (e.g., to form a lens that aligns with eye box **24**). Other mounting arrangements may be used, if desired.

(28) As shown in FIG. **2**, display module **14A** may generate light **22** associated with image content to be displayed to eye box **24**. Light **22** may be collimated using a lens such as collimating lens **34**. Optical system **14B** may be used to present light **22** output from display module **14A** to eye box **24**.

(29) Optical system **14B** may include one or more optical couplers such as input coupler **28**, cross-coupler **32**, and output coupler **30**. In the example of FIG. **2**, input coupler **28**, cross-coupler **32**, and output coupler **30** are formed at or on waveguide **26**. Input coupler **28**, cross-coupler **32**, and/or output coupler **30** may be completely embedded within the substrate layers of waveguide **26**, may be partially embedded within the substrate layers of waveguide **26**, may be mounted to waveguide **26** (e.g., mounted to an exterior surface of waveguide **26**), etc.

(30) The example of FIG. **2** is merely illustrative. One or more of these couplers (e.g., cross-coupler **32**) may be omitted. Optical system **14B** may include multiple waveguides that are laterally and/or vertically stacked with respect to each other. Each waveguide may include one, two, all, or none of couplers **28**, **32**, and **30**. Waveguide **26** may be at least partially curved or bent if desired.

(31) Waveguide **26** may guide light **22** down its length via total internal reflection. Input coupler **28** may be configured to couple light **22** from display module **14A** (lens **34**) into waveguide **26**, whereas output coupler **30** may be configured to couple light **22** from within waveguide **26** to the exterior of waveguide **26** and towards eye box **24**. For example, display module **14A** may emit light **22** in direction +Y towards optical system **14B**. When light **22** strikes input coupler **28**, input coupler **28** may redirect light **22** so that the light propagates within waveguide **26** via total internal reflection towards output coupler **30** (e.g., in direction X). When light **22** strikes output coupler **30**,

output coupler **30** may redirect light **22** out of waveguide **26** towards eye box **24** (e.g., back along the Y-axis). In scenarios where cross-coupler **32** is formed at waveguide **26**, cross-coupler **32** may redirect light **22** in one or more directions as it propagates down the length of waveguide **26**, for example.

(32) Input coupler **28**, cross-coupler **32**, and/or output coupler **30** may be based on reflective and refractive optics or may be based on holographic (e.g., diffractive) optics. In arrangements where couplers **28**, **30**, and **32** are formed from reflective and refractive optics, couplers **28**, **30**, and **32** may include one or more reflectors (e.g., an array of micromirrors, partial mirrors, or other reflectors). In arrangements where couplers **28**, **30**, and **32** are based on holographic optics, couplers **28**, **30**, and **32** may include diffractive gratings (e.g., volume holograms, surface relief gratings, etc.).

(33) In one suitable arrangement that is sometimes described herein as an example, output coupler **30** is formed from diffractive gratings or micromirrors embedded within waveguide **26** (e.g., volume holograms recorded on a grating medium stacked between transparent polymer waveguide substrates, an array of micromirrors embedded in a polymer layer interposed between transparent polymer waveguide substrates, etc.), whereas input coupler **28** is formed from a reflective prism mounted to an exterior surface of waveguide **26** (e.g., an exterior surface defined by a waveguide substrate that contacts the grating medium or the polymer layer used to form output coupler **30**).

(34) FIG. **3** is a top view showing how input coupler **28** at waveguide **26** may be formed from a reflective prism. As shown in FIG. **3**, input coupler **28** may include a reflective prism (e.g., a reflective input coupling prism) such as prism **36**. Prism **36** may have a bottom surface **38** mounted to exterior surface **40** of waveguide **26** (e.g., using an optically clear adhesive not shown in FIG. **3** for the sake of clarity).

(35) As shown in FIG. **3**, lens **34** may receive light **22** (e.g., from display module **14A** of FIG. **2**). Prism **36** may be mounted to the side of waveguide **26** opposite to lens **34** and display module **14A**. For example, waveguide **26** may have an exterior surface **42** that opposes exterior surface **40**. Exterior surface **42** may be interposed between prism **36** and lens **34** (e.g., waveguide **26** may be interposed between prism **36** and display module **14A**).

(36) Lens **34** may direct light **22** towards waveguide **26**. Light **22** may enter waveguide **26** through surface **42** (e.g., at a sufficiently low angle with respect to the normal surface of surface **42** such that no total internal reflection occurs). Light **22** may pass through surface **40** of waveguide **26** into prism **36**. Light **22** may reflect off of surface **44** of prism **36** (sometimes referred to herein as reflective surface **44** or reflection surface **44**) and back into waveguide **26** through surfaces **38** and **40**. Surface **44** may be tilted in one or more directions (e.g., including out of the plane of the page, where the normal axis of surface **44** is oriented at a non-zero angle with respect to the +Y axis within the Z-Y plane in addition to a non-zero angle with respect to the +Y axis within the X-Y plane). Light **22** may then propagate down the length of waveguide **26** via total internal reflection. In this way, prism **36** may couple light **22** into waveguide **26** despite being located on the side of waveguide **26** opposite to lens **34** and display module **14A** (FIG. **2**).

(37) The angular spread of light **22** may be confined to pupil **46** within waveguide **26** after being coupled into waveguide **26** by prism **36** (e.g., the rays of light reflected by prism **36** may converge at pupil **46** within waveguide **26**). The optical path length from surface **42** to surface **38**, from surface **38** to surface **44**, and to pupil **46** from surface **44** may, for example, be substantially greater (e.g., two times greater or more than two times greater) than the optical path length from lens **34** to the point within the waveguide where the rays of light from the prism converge in scenarios where input coupler **28** is formed from a transmissive input coupling prism mounted to surface **42** of waveguide **26**. This may serve to optimize the optical performance of optical system **14B** while minimizing the distance **53** from the input of lens **34** to surface **42** (e.g., thereby allowing optical system **14B** to occupy less space and to better fit within the form factor of a head mounted support structure such as support structure **20** of FIG. **1**) relative to scenarios where a transmissive input

coupling prism is used.

(38) Lens **34** may be formed from one or more lens elements **48** (e.g., a first lens element **48-1**, an Nth lens element **48-N**, etc.). Lens elements **48** may include any desired lens elements (e.g., lenses with concave surfaces, convex surfaces, planar surfaces, spherical surfaces, aspherical surfaces, and/or free-form curved surfaces, lenses with microstructures, Fresnel lenses, combinations of these and/or other types of lens elements, etc.). Lens **34** may have a width **52** and a length **50**.

(39) If desired, an optional reflective layer such as reflective layer **45** may be layered onto surface **44** of prism **36**. Reflective layer **45** may include a reflective dielectric coating, a reflective metallic coating, patterns of three-dimensional metal structures, and/or a layer of (reflective) diffraction gratings such as one or more thin-film or volume holograms in a grating medium or surface relief gratings layered onto surface **44**. Reflective layer **45** may, for example, help to increase the amount of light **22** that is reflected off of surface **44**. If desired, reflective layer **45** may also be configured to provide light **22** with an optical power upon reflecting off of reflective layer **45** at surface **44** of prism **36**. For example, diffractive gratings in reflective layer **45** may diffract light **22** back towards waveguide **26** while also imparting optical power onto the diffracted light. Reflective layer **45** may be omitted if desired.

(40) In the example of FIG. 3, surface **44** of prism **36** is planar. This example is merely illustrative. If desired, surface **44** may be curved. FIG. 4 is a top view showing how surface **44** of prism **36** may be curved. As shown in FIG. 4, the curved shape of surface **44** may impart light **22** with optical power upon reflection off of surface **44**. Surface **44** may have a spherically-curved shape, an aspherically-curved shape, or a free-form curved shape following any desired curved three-dimensional surface. Reflective layer **45** on the curved surface **44** of FIG. 4 may help to increase the amount of light **22** that is reflected off of surface **44** and/or may help contribute to the optical power of the light **22** reflected into waveguide **26**. Reflective layer **45** may be omitted if desired.

(41) In the example of FIG. 4, lens **34** includes M lens elements **48** (e.g., a first lens element **48-1**, an Mth lens element **48-M**, etc.). The number M of lens elements **48** in lens **34** in the example of FIG. 4 may be less than the number N of lens elements **48** in lens **34** in the example of FIG. 3. The optical power imparted by curved surface **44** and/or reflective layer **45** may, for example, contribute the optical power to light **22** that would otherwise have been provided by the N-M extra lenses in lens **34** of FIG. 3. This may allow the input of lens **34** to be separated from surface **42** of waveguide **26** by a distance **54** that is shorter than distance **53** of FIG. 3. This may serve to further reduce the amount of space occupied by optical system **14B** in system **10** (FIG. 1), for example.

(42) In the example of FIGS. 3 and 4, light **22** may be confined to a relatively small pupil within waveguide **26** (e.g., pupil **46** of FIG. 3). In practice, it may be desirable for optical system **14B** to fill as large of an eye box **24** with as uniform an intensity of light **22** as possible. If desired, light redirecting elements such as input coupler **28** may be configured to expand light **22** in one or more dimensions while also coupling light **22** into waveguide **26**, to fill as large of an eye box **24** with as uniform an intensity of light **22** as possible.

(43) FIG. 5 shows an example in which prism **36** includes a switchable reflective surface for expanding the field of view of light **22**. As shown in FIG. 5, a switchable reflective layer such as switchable reflective layer **56** may be layered onto surface **44** of prism **36**. Switchable reflective layer **56** may include, for example, a liquid crystal (LC) grating, a digital micromirror device (DMD), or other microelectromechanical (MEMs) structures that are switchable between at least first and second states. Switchable reflective layer **56** may receive control signals over control path **58** (e.g., from control circuitry **16** of FIG. 1) that place switchable reflective layer **56** into a selected one of the first and second states at any given time.

(44) In the first state, image light **22** may reflect off of surface **44** and switchable reflective layer **56** (sometimes referred to herein collectively as a switchable reflective surface) in a first direction into waveguide **26**, as shown by arrow (ray) **60**. This reflected light may exhibit a relatively-small field of view **62**. In the second state, image light **22** may reflect off of surface **44** and switchable

reflective layer **56** in a second direction into waveguide **26**, as shown by arrow **64**. This reflected light may exhibit a relatively-small field of view **66**.

(45) The control circuitry may rapidly toggle switchable reflective layer **56** between the first and second states, as shown by arrow **68**, so that the light **22** coupled into waveguide **26** exhibits a relatively wide effective field of view **70**. Switchable reflective layer **56** may be switched between the first and second states at a speed greater than the response speed of the human eye (e.g., greater than 60 Hz, greater than 120 Hz, greater than 240 Hz, greater than 1 kHz, greater than 10 kHz, etc.) so that a user at eye box **24** (FIG. 2) is unable to separately perceive each state and instead perceives a single effective field of view **70**. In this way, image light **22** may be coupled into waveguide **26** and provided to the eye box with a wider effective field of view than would otherwise be provided to the eye box. As an example, fields of view **62** and **66** may each be 30 degrees, 25 degrees, between 25 and 35 degrees, less than 45 degrees, etc., whereas field of view **70** is 60 degrees, between 55 and 65 degrees, greater than 45 degrees, or any other desired angle greater than field of view **62** or field of view **66**.

(46) FIG. 6 shows another example in which prism **36** expands the field of view of light **22**. As shown in FIG. 6, prism **36** may be a split prism having a first portion **76** and a second portion **78** stacked on upper surface **72** of first portion **76**. Upper surface **72** may be planar, may be tilted in one or more directions (e.g., including out of the plane of the page), or may be curved (e.g., free-form curved). Portions **78** and **76** may be formed from the same material (e.g., may have the same index of refraction) or may be formed from different materials (e.g., materials having different indices of refraction).

(47) Portion **76** may have a wedge shape and may therefore sometimes be referred to herein as wedge **76**. Portion **78** may have a wedge shape and may therefore sometimes be referred to herein as wedge **78**. This is merely illustrative and, in general, portions **76** and **78** may have any desired shapes. Wedge **76** may include bottom surface **38** of prism **36** and may be mounted to surface **40** of waveguide **26**. Upper surface **72** of wedge **76** may be oriented at an acute angle such as angle **80** with respect to surface **38** (e.g., surfaces **38**, **40**, and/or **42** may be planar). Wedge **78** may include the reflective surface **44** of prism **36**. Surface **44** may be oriented at an acute angle such as angle **82** with respect to surface **80** (e.g., angle **82** may be greater than angle **80**).

(48) Prism **36** may include a passive partially reflective layer such as polarization-sensitive reflective film **74**. Polarization-sensitive reflective film **74** (sometimes referred to herein as polarizing beam splitter **74** or reflective film **74**) may be layered onto upper surface **72** of wedge **76** (e.g., wedge **78** may be mounted to reflective film **74**). Reflective film **74** may reflect light of a first polarization and may pass (transmit) light of a second polarization without reflecting the light of the second polarization. The first polarization may be a p-polarization whereas the second polarization is an s-polarization or the first polarization may be an s-polarization whereas the second polarization is a p-polarization, as an example. If desired, optically clear adhesive (not shown in FIG. 6 for the sake of clarity) may be used to help adhere wedge **78** to wedge **76**. Reflective film **74** may be formed from a wire grid polarizer, if desired.

(49) In this example, optical system **14B** may also include a switchable polarizer such as switchable polarizer **84**. Waveguide **26** may be interposed between switchable polarizer **84** and prism **36**. Switchable polarizer **84** may be mounted to surface **42** of waveguide **26** (e.g., using an optically clear adhesive not shown in FIG. 6 for the sake of clarity) or may be spaced apart from surface **42**. One or more lens elements (e.g., from lens **34** of FIG. 2) may be interposed between switchable polarizer **84** and waveguide **26**. In another suitable arrangement, display module **14A** may include a spatial light modulator such as a reflective display panel (e.g., a DMD or LCOS display panel) or a transmissive display panel that receives illumination light from one or more light sources. In this example, switchable polarizer **84** may, if desired, be optically coupled between the light sources and the spatial light modulator.

(50) Switchable polarizer **84** may receive light **22**. Switchable polarizer **84** may have a first state at

which switchable polarizer **84** only passes (transmits) light **22A** of the first polarization and a second state at which switchable polarizer **84** only passes (transmits) light **22B** of the second polarization. When switchable polarizer **84** is in the first state, light **22A** of the first polarization passes through wedge **76** and reflects off of reflective film **72** to produce field of view **62** within waveguide **26**. When switchable polarizer **84** is in the second state, light **22B** of the second polarization passes through wedge **76** and reflective film **74** (e.g., without being reflected by reflective film **74**). Light **22B** then passes through wedge **78** and is reflected off of surface **44** of wedge **78** to produce field of view **66** within waveguide **26**. If desired, reflective layer **45** may be formed on surface **44** to help increase the reflected intensity and/or to provide optical power to light **22B**. Surface **44** and/or surface **72** (e.g., reflective film **74**) may be curved if desired (e.g., to provide light **22A** and **22B** with optical power). Reflective layer **45** may be omitted if desired.

(51) Switchable polarizer **84** may receive control signals over control path **86** (e.g., from control circuitry **16** of FIG. **1**) that place switchable polarizer **84** into a selected one of the first and second states at any given time. The control circuitry may rapidly toggle switchable polarizer **84** between the first and second states, as shown by arrow **68**, so that light **22** (e.g., light **22A** and **22B**) is coupled into waveguide **26** and exhibits the relatively wide effective field of view **70**. Switchable polarizer **84** may be switched between the first and second states at a speed greater than the response speed of the human eye (e.g., greater than 60 Hz, greater than 120 Hz, greater than 240 Hz, greater than 1 kHz, greater than 10 kHz, etc.) so that a user at eye box **24** (FIG. **1**) is unable to perceive each state and instead perceives a single effective field of view **70**. In this way, image light **22** may be coupled into waveguide **26** and provided to the eye box with a wider effective field of view than would otherwise be provided to the eye box. In another suitable arrangement, rapidly toggling between light **22A** and **22B** may be used to increase the effective resolution of the images provided at the eye box (e.g., where the light reflected by surfaces **72** and **44** is angularly separated by about one-half of the pixel pitch of the image data to be displayed, thereby doubling the effective resolution of the images provided to the eye box). The methods and systems for increasing field of view as described in connection with any of the embodiments herein may additionally or alternatively be used to increase the effective resolution of the images provided at the eye box in this way (e.g., the arrangement of FIG. **5** may also be used to produce an image having an increased effective resolution at the eye box rather than to expand the field of view of the light at the eye box).

(52) FIG. **7** shows another example in which prism **36** expands the field of view of light **22**. As shown in FIG. **7**, prism **36** may include a switchable reflective layer such as switchable reflective layer **88** interposed between wedges **76** and **78**. Switchable reflective layer **88** may include a cholesteric liquid crystal reflector or any other desired switchable reflective (mirror) layered onto surface **72** of wedge **76** (e.g., wedge **78** may be mounted to switchable reflective layer **88**).

(53) Switchable reflective layer **88** may receive light **22** through waveguide **26** and wedge **76**. Switchable reflective layer **88** may have a first state at which switchable reflective layer **88** reflects light **22** and a second state at which switchable reflective layer **88** only passes (transmits) light **22** without reflecting light **22**. When switchable reflective layer **88** is in the first state, light **22** passes through wedge **76** and reflects off of switchable reflective layer **88** to produce field of view **62** within waveguide **26**. When switchable reflective layer **88** is in the second state, light **22** passes through wedge **76** and switchable reflective layer **88** (e.g., without being reflected by switchable reflective layer **88**). Light **22** then passes through wedge **78** and is reflected off of surface **44** of wedge **78** to produce field of view **66** within waveguide **26**. If desired, reflective layer **45** may be formed on surface **44** to help increase the reflected intensity and/or to provide optical power to light **22**. Surface **44** and/or surface **72** (e.g., switchable reflective layer **88**) may be curved if desired (e.g., to provide light **22** with optical power). Reflective layer **45** may be omitted if desired.

(54) Switchable reflective layer **88** may receive control signals over control path **90** (e.g., from control circuitry **16** of FIG. **1**) that place switchable reflective layer **88** into a selected one of the

first and second states at any given time. The control circuitry may rapidly toggle switchable reflective layer **88** between the first and second states, as shown by arrow **68**, so that light **22** is coupled into waveguide **26** and exhibits the relatively wide effective field of view **70**. Switchable reflective layer **88** may be switched between the first and second states at a speed greater than the response speed of the human eye (e.g., greater than 60 Hz, greater than 120 Hz, greater than 240 Hz, greater than 1 kHz, greater than 10 kHz, etc.) so that a user at eye box **24** (FIG. 1) is unable to perceive each state and instead perceives a single effective field of view **70**. In this way, image light **22** may be coupled into waveguide **26** and provided to the eye box with a wider effective field of view than would otherwise be provided to the eye box. Because switchable reflective layer **88** is insensitive to the polarization of light **22**, prism **36** of FIG. 7 may expand the field of view of light **22** without requiring other components that polarize light **22**.

(55) In the example of FIGS. 6 and 7, light **22** is expanded by toggling a switchable element faster than the response speed of the human eye (e.g., because image light is only provided within one of fields of view **62** or **66** at any given time). To accommodate this switching, display module **14A** needs to produce frames of image data in light **22** at the same relatively high speed, so that a full stream of image data is provided to a user at the eye box. In practice, this can consume an undesirably high amount of processing resources and power within system **10**.

(56) In another suitable arrangement, light **22** may be concurrently provided within both fields of view **62** and **66** in waveguide **26**. This may allow the field of view of light **22** to be expanded without rapidly toggling a switchable element and without requiring the display module to display image data at a corresponding high speed. This may serve to minimize the amount of processing resources and power consumed by system **10** while still expanding the field of view of light **22** provided to the eye box. In another suitable arrangement, the structures of FIG. 7 may be used to produce an image having an increased effective resolution at the eye box rather than to expand the field of view of the light at the eye box).

(57) FIG. 8 is a diagram showing how light **22** may be concurrently provided within both fields of view **62** and **66** in waveguide **26**. As shown in FIG. 8, display module **14A** may include at least a first display panel **93** and a second display panel **95**. Display module **14A** may also include a prism such as prism **94**. Display panels **93** and **95** may be, for example, uLED panels or other emissive displays. Collimating optics, polarizers, and/or any other desired optical components may be interposed between display panel **93** and prism **94** and between display panel **95** and prism **94**. Display panels **93** and **95** may include reflective spatial light modulators (e.g., DMD panels, LCOS panels, etc.) in another suitable arrangement.

(58) Display panel **93** may emit light **22A** of the first polarization whereas display panel **95** emits light **22B** of the second polarization. Prism **94** may include a polarizing beam splitter that reflects light **22A** of the first polarization towards lens **34** and input coupler **28** and that transmits light **22B** of the second polarization towards lens **34** and input coupler **28**. Reflective film **74** on wedge **76** may reflect light **22A** to produce field of view **62** within waveguide **26**. At the same time, reflective film **74** may transmit light **22B**, which reflects off of surface **44** to produce field of view **66** within waveguide **26**. Display panels **93** and **95** may be active at the same time to concurrently provide light **22A** and **22B** to input coupler **28**. This may allow prism **36** to concurrently produce fields of view **62** and **66**, so that the in-coupled light collectively exhibits the wide effective field of view **70** within waveguide **26**.

(59) The pupils of light **22** within waveguide **26** (e.g., pupil **46** of FIG. 3) may be replicated during propagation down the length of waveguide **26** to fill the eye box with light. In practice, there may be gaps between the replicated pupils that produce uneven brightness uniformity across the area of the eye box. In some scenarios, increasing the size of display module **14A** may help to fill the gaps between the replicated pupils so that the eye box is filled with light of uniform intensity. However, larger display modules **14A** may occupy an excessive amount of space within system **10**. If desired, the optical path of optical system **14B** may include a birefringent beam displacer that helps to fill

the gaps between the replicated pupils so that the eye box is filled with light of uniform intensity (e.g., without increasing the size of display module **14A**). In another suitable arrangement, the structures of FIG. **8** may be used to produce an image having an increased effective resolution at the eye box.

(60) FIG. **9** is a perspective view of an illustrative birefringent beam displacer **102** that may be provided in optical system **14B**. As shown in FIG. **9**, birefringent beam displacer **102** may be formed from a birefringent material such as calcite and may have a length **100** (e.g., in the direction of the optical path). Birefringent beam displacer **102** may receive light **22**. Light **22** may include light of a first polarization **96** in alignment with light of a second polarization **98** (e.g., within a corresponding pupil **104** at the input face of birefringent beam displacer **102**).

(61) Birefringent beam displacer **102** may separate light **22** into a first beam **108** that includes the light of the first polarization **96** and a second beam **109** that includes the light of the second polarization **98**. Upon exiting birefringent beam displacer **102**, beam **109** may be separated from beam **108** by displacement **107**. The magnitude of displacement **107** may be directly proportional to the length **100** of birefringent beam displacer **102**, for example. Beams **108** and **109** may each individually exhibit relatively small pupils **104**. However, displacement **107** may separate the pupils **104** of beams **108** and **109** so that beams **108** and **109** collectively exhibit an expanded pupil **106**. Expanded pupil **106** may be replicated by waveguide **26**. Because expanded pupil **106** is larger than pupil **104**, expanded pupil **106** may fill the eye box with light of more uniform intensity relative to scenarios where the birefringent beam displacer is omitted, for example.

(62) FIG. **10** shows one example of how birefringent beam displacer **102** may be mounted within optical system **14B**. As shown in FIG. **10**, waveguide **26** may be interposed between prism **36** and birefringent beam displacer **102**. Birefringent beam displacer **102** may be mounted to surface **42** (e.g., using optically clear adhesive not shown in FIG. **10** for the sake of clarity) or may be separated from waveguide **26**.

(63) Wedge **76** of prism **36** may, for example, be formed from a material having a first index of refraction $n_{\text{sub.1}}$ whereas wedge **78** is formed from a material having a second index of refraction $n_{\text{sub.2}}$. The difference in indices $n_{\text{sub.1}}$ and $n_{\text{sub.2}}$ may cause the light (e.g., after being split into displaced beams **108** and **109** of FIG. **9**) to reflect off of surface **72** so that the light (e.g., expanded pupil **106** of FIG. **9**) is coupled into waveguide **26**. Partially or completely reflective layers may, if desired, be formed on any of the surfaces of prism **36** shown in FIG. **10**, which may be planar and/or curved. In another suitable arrangement, birefringent beam displacer **102** may be used to form some or all of prism **36** (e.g., wedges **76** and/or **78**). In another suitable arrangement, the structures of FIG. **10** may be used to produce an image having an increased effective resolution at the eye box.

(64) In one suitable arrangement, lens **34** of FIGS. **2-4** may include a single set or grouping of lens elements (e.g., lens elements **48** of FIGS. **3** and **4**) that direct light **22** towards waveguide **26** without converging the light at a point between two or more of the lens elements. This is merely illustrative. If desired, lens **34** may be provided with a relay optics arrangement in which lens **34** is formed from multiple groups or sets of lens elements that are separated by a gap, where the lens elements converge light **22** at a point between two or more of the lens elements.

(65) FIG. **11** is a diagram showing how lens **34** may include multiple groups or sets of lens elements that are separated by a gap, where the lens elements converge light **22** at a point between two or more of the lens elements. As shown in FIG. **11**, lens **34** may include a first set (group) of lens elements **34A** and a second set (group) of lens elements **34B** (e.g., sets of lens elements such as lens elements **48** of FIGS. **3** and **4**). Lens elements **34A** and lens elements **34B** may each include any desired number of lens elements (e.g., one lens element, two lens elements, three lens elements, four lens elements, more than four lens elements, etc.). Lens **34** may include the same number of lens elements **34A** as lens elements **34B** or may include a different number of lens elements **34A** and **34B**. Lens elements **34A** and **34B** may include any desired types of lens

elements. If desired, physical light stops such as optical absorbers **114** may be placed at one or more locations in lens **34** (e.g., in gap **110**, around lens elements **34A**, etc.) to help mitigate stray light in the system.

(66) Lens elements **34A** may be separated from lens elements **34B** in lens **34** by gap **110**. Gap **110** may be greater than the distance between adjacent lens elements **34B** and greater than the distance between adjacent lens elements **34A**, for example. Lens elements **34A** may receive light **22** from display module **14A**. Lens elements **34A** may focus (converge) light **22** at point (pupil) **112** between lens elements **34A** and lens elements **34B**. The rays of light **22** may then reach lens elements **34B**, which redirect the light towards waveguide **26** (e.g., surface **42** of waveguide **26** as shown in FIGS. **3-8** and **10**). Arranging the lens elements in lens **34** in this way may allow lens **34** to exhibit a length **116** and a width **118**. Length **116** of lens **34** in FIG. **11** may, for example, be greater than the length **50** of lens **34** in scenarios where only a single group of lens elements is used (e.g., as shown in FIG. **3**). However, width **118** may be less than the width **52** of lens **34** in scenarios where only a single group of lens elements is used (e.g., as shown in FIG. **3**). This may, for example, reduce the amount of space required within device **10** of FIG. **1** without sacrificing image quality. The example of FIG. **11** is merely illustrative and, if desired, other lens element arrangements may be used.

(67) If desired, a switchable element may be mounted within gap **110** of lens **34** to help increase the effective resolution of display **14**. FIG. **12** is a diagram showing how a switchable element may be mounted within gap **110** of lens **34** to help increase the effective resolution of display **14**. As shown in FIG. **12**, a switchable element such as switchable light displacing element **120** may be mounted at point **112** within gap **110**. Switchable light displacing element **120** may be a flat transparent plate (e.g., a glass plate, plastic plate, etc.) that is switchable between first and second states.

(68) In the first state, switchable light displacing element **120** may have a first orientation. In the second state, switchable light displacing element **120** may have a second orientation. Switchable light displacing element **120** may include MEMs structures, piezoelectric structures, or other structures for adjusting the orientation of the switchable element so that the switchable element is placed in a selected one of the first and second states. Switchable light displacing element **120** may receive control signals over control path **122** (e.g., from control circuitry **16** of FIG. **1**) that place switchable light displacing element **120** into a selected one of the first and second states at any given time.

(69) FIG. **13** is a diagram showing how switchable light displacing element **120** may be placed into the first and second states. As shown in FIG. **13**, switchable light displacing element **120** may have a first orientation such as first orientation **124** in the first state. Switchable light displacing element **120** may have a second orientation such as second orientation **126** in the second state. Light **22** from a single pixel in display module **14A** is shown in the example of FIG. **13** for the sake of clarity. In general, light **22** is received at switchable light displacing element **120** for each pixel in the display module.

(70) In the first state, light **22** may pass directly through switchable light displacing element **120**, as shown by ray **128**. In the first state, light **22** may be incident upon switchable light displacing element **120** parallel to a normal axis of the lateral area of switchable light displacing element **120**. This may allow light **22** to pass through switchable light displacing element **120** without refracting. In the second state, light **22** may be incident upon switchable light displacing element **120** at a non-zero angle with respect to the normal axis of switchable light displacing element **120**. This may cause light **22** to refract prior to being output by switchable light displacing element **120**, as shown by ray **130**. This may cause light **22** (ray **130**) to be displaced in the second state by an offset such as offset **132** relative to the first state (ray **128**).

(71) The control circuitry may rapidly toggle (e.g., rotate) switchable light displacing element **120** between the first and second states, as shown by arrow **134**, so that the light **22** is provided to the waveguide with a first alignment, as shown by ray **128**, and a second alignment (e.g., a second

alignment that is parallel to and offset from the first alignment), as shown by ray **130**. Similar rays are produced for each pixel in the display, such that the first and second alignments collectively cause the light to exhibit a greater resolution than would otherwise be available in the absence of switchable light displacing element **120**. Switchable light displacing element **120** may be switched between the first and second states at a speed greater than the response speed of the human eye (e.g., greater than 60 Hz, greater than 120 Hz, greater than 240 Hz, greater than 1 kHz, greater than 10 kHz, etc.) so that a user at eye box **24** (FIG. **2**) is unable to perceive each state. In this way, light **22** may be coupled into waveguide **26** and provided to the eye box with a greater effective resolution than would otherwise be provided to the eye box in the absence of switchable light displacing element **120**.

(72) FIG. **14** is a front view showing how switchable light displacing element **120** may provide light **22** with an increased effective resolution to the waveguide (e.g., as taken in the direction of arrow **131** of FIG. **14**). In the example of FIG. **14**, four pixels of light **22** are shown for the sake of clarity. In general, light **22** and the display module may include any desired number of pixels.

(73) As shown in FIG. **14**, light **22** may include pixels **P1**, **P2**, **P3**, and **P4** when switchable light displacing element **120** is in the first state (e.g., orientation **124** of FIG. **13**). When switchable light displacing element **120** is in the second state (e.g., orientation **126** of FIG. **13**), pixels **P1**, **P2**, **P3**, and **P4** may be displaced by displacement **138**, as shown by respective pixels **P1'**, **P2'**, **P3'**, and **P4'**. Displacement **138** may, for example, be a two-dimensional displacement that includes offset **136** parallel to the Z-axis and/or offset **132** parallel to the X-axis.

(74) Pixels **P1**, **P2**, **P3**, and **P4** may exhibit a first pixel pitch and pixels **P1'**, **P2'**, **P3'**, and **P4'** may exhibit the first pixel pitch. However, the combination of pixels **P1**, **P2**, **P3**, and **P4** with pixels **P1'**, **P2'**, **P3'**, and **P4'** may exhibit a second pixel pitch that is less than (e.g., half) the first pixel pitch. By rapidly toggling between the first and second states, light **22** may effectively include each of pixels **P1**, **P2**, **P3**, **P4**, **P1'**, **P2'**, **P3'**, and **P4'** (e.g., as perceived by a user at the eye box) and thus the second pixel pitch, rather than only pixels **P1**, **P2**, **P3**, and **P4** and the first pixel pitch (e.g., in scenarios where switchable light displacing element **120** is omitted). This may serve to increase the effective resolution of light **22** relative to scenarios where switchable light displacing element **120** is omitted (e.g., to twice the resolution that light **22** would otherwise have in the absence of switchable light displacing element **120**), without requiring an increase in size or processing resources for display module **14A**.

(75) In scenarios where prism **36** includes a curved reflective surface (e.g., surface **44** of FIG. **4**), the optical power introduced by the curved reflective surface may allow lens elements **34B** of lens **34** (FIGS. **11** and **12**) to be omitted. FIG. **15** is a diagram showing how the curved reflective surface of prism **36** may replace lens elements **34B** of FIGS. **11** and **12**.

(76) As shown in FIG. **15**, lens elements **34A** of lens **34** may converge light **22** at point **112**. Light **22** may subsequently pass through waveguide **26** to prism **36** and may be reflected off of the curved surface **44** of prism **36**. The curve of surface **44** may introduce optical power to the light that would otherwise have been introduced by lens elements **34B** of FIGS. **11** and **12**. This may allow lens elements **34B** to be omitted, allowing lens elements **34A** to be placed closer to waveguide **26** than in scenarios where lens elements **34B** are included in lens **34**. For example, there may be a distance **140** between the input to lens **34** and surface **42** of waveguide **26**, where distance **140** is less than length **116** of FIG. **11** and distance **53** of FIG. **3**. Switchable light displacing element **120** may be mounted at point **112** for increasing the effective resolution of light **22** if desired.

(77) The example of FIG. **15** is merely illustrative. If desired, reflective layer **45** may be layered onto surface **44** or may be omitted. Switchable light displacing element **120** may be omitted if desired. Prism **36** of FIG. **15** may include one or more wedges such as wedges **76** and **78** of FIGS. **6-8** and **10**). Surface **44** may have any desired free-form curved shape. Any desired combination of the arrangements of FIGS. **3-15** may be used.

(78) In practice, display module **14A** and lens **34** may focus image light **22** onto the reflective surface(s) of prism **36** within a corresponding pupil. If desired, the structures layered onto the reflective surface(s) of prism **36** may extend over (overlap) only a portion of the reflective surface(s), where the portion overlaps the pupil of the image light **22** focused on the reflective surface(s) (e.g., the portion may have the same size and shape as the pupil or may be slightly larger than the pupil). The remainder of the reflective surface(s) may be covered (overlapped) by an optical absorber that serves as a physical stop for the image light. For example, as shown in FIG. **16**, prism **36** may include reflective layer **150** on reflective surface **44** that reflects image light **22** into waveguide **26**. The size of reflective layer **150** may match the pupil size of image light **22**. The remainder of reflective surface **44** may be covered by optical absorber **152**. This example is merely illustrative and, in general, reflective layer **45** of FIGS. **3**, **4**, **6-8**, and **15**, switchable reflective layer **56** of FIG. **5**, polarization-sensitive reflective film **74** of FIGS. **6** and **8**, and switchable reflective layer **88** of FIG. **7** may overlap only a portion of the corresponding reflective surface of prism **36** that overlaps (e.g., matches) the area spanned by the pupil of image light **22** focused on the reflective surface (e.g., as shown by reflective layer **150** of FIG. **16**), whereas the remainder of the reflective surface is covered with an optical absorber (e.g., as shown by optical absorber **152** of FIG. **16**). This may serve to minimize stray light in optical system **14B**, thereby maximizing the contrast of images provided at eye box **24**.

(79) If desired, an optional optical wedge such may be interposed on the optical path between waveguide **26** and display modules **14A**, as shown by optical wedge **158** of FIG. **16**. Optical wedge **158** may have a first surface **154** facing and parallel to the lateral surface of waveguide **26**. Optical wedge **158** may have an opposing second surface **156** that is oriented at a non-parallel angle (e.g., tilted) with respect to first surface **154**. Image light **22** may pass through optical wedge **158** before passing through waveguide **26**. Optical wedge **158** may be a dispersion compensation wedge that compensates for dispersion of image light **22** by prism **36** and/or the other optical components of the system (e.g., in scenarios where prism **36** has a refractive index as a function of wavelength that is different from the bulk refractive index as a function of wavelength of the grating medium in waveguide **26**). If desired, optical wedge **158** may additionally or alternatively help redirect image light **22** incident at other angles (e.g., angles non-parallel with respect to the Y-axis) towards prism **36**. This may allow display module **14A** to be mounted at different locations or orientations with respect to waveguide **26** than would otherwise be possible in the absence of the optical wedge. For example, the optical wedge may allow display module **14A** to be located within a main frame for waveguide **26** (e.g., within support structures **20** of FIG. **1**) without needing to be located in the temple or other portions of the support structures (e.g., thereby optimizing space consumption within system **10**). Optical wedge **158** of FIG. **16** may be interposed on the optical path for image light **22** between waveguide **26** and display module **14A** in any of the arrangements of FIGS. **3-8**, **10**, and **15** if desired.

(80) In accordance with an embodiment, a display system is provided that includes a display module that produces image light, a waveguide, a reflective input coupling prism mounted to the waveguide, the reflective input coupling prism is configured to receive the image light through the waveguide, the reflective input coupling prism is configured to reflect the image light into the waveguide, and the waveguide is configured to propagate the reflected image light via total internal reflection, and an output coupler on the waveguide and configured to couple the reflected image light out of the waveguide.

(81) In accordance with another embodiment, the output coupler includes diffractive grating structures in a grating medium embedded within the waveguide, the diffractive grating structures include volume holograms, and the display system includes a dispersion compensation wedge optically interposed between the display module and the waveguide.

(82) In accordance with another embodiment, the reflective input coupling prism has a reflective surface configured to reflect the image light into the waveguide, the display system includes a

reflective layer on the reflective surface.

(83) In accordance with another embodiment, the reflective layer includes a switchable reflective layer, the switchable reflective layer being adjustable between a first state at which the switchable reflective layer reflects the image light into the waveguide in a first direction and a second state at which the switchable reflective layer reflects the image light into the waveguide in a second direction that is different from the first direction, and the switchable reflective layer includes a structure selected from the group consisting of a switchable liquid crystal grating and a digital micromirror device.

(84) In accordance with another embodiment, the reflective layer includes a layer selected from the group consisting of a dielectric layer, a metallic layer, a layer of three-dimensional metal structures, and a layer of diffractive gratings.

(85) In accordance with another embodiment, the reflective surface includes a curved surface that is configured to provide the reflected light with an optical power.

(86) In accordance with another embodiment, the reflective layer covers a portion of the reflective surface that overlaps a pupil of the image light that has been focused onto the reflective surface and the display system includes an optical absorber that covers a remainder of the reflective surface.

(87) In accordance with another embodiment, the display includes a lens configured to direct the image light from the display module towards the waveguide, the lens is configured to converge the image light at a location interposed between the waveguide and the lens, and a transparent plate at the location, the transparent plate being adjustable between a first orientation at which the transparent plate directs the image light towards the waveguide with a first alignment and a second orientation at which the transparent plate directs the image light towards the waveguide with a second alignment that is parallel to and offset from the first alignment.

(88) In accordance with another embodiment, the waveguide has opposing first and second surfaces, the reflective input coupling prism includes a first wedge mounted to the first surface and a second wedge mounted to a surface of the first wedge, the second wedge includes the reflective surface, the surface of the first wedge is configured to reflect the image light into the waveguide in a first direction, and the reflective surface is configured to reflect the image light into the waveguide in a second direction that is different from the first direction.

(89) In accordance with another embodiment, the reflective input coupling prism has a reflective surface configured to reflect the image light into the waveguide, the display includes a polarization-sensitive film interposed between the first and second wedges, the polarization-sensitive film is configured to reflect a first polarization of the image light in the first direction and to transmit a second polarization of the image light, and the reflective surface is configured to reflect, in the second direction, the second polarization of the image light transmitted by the polarization-sensitive film.

(90) In accordance with another embodiment, the display includes a switchable polarizer, the switchable polarizer is switchable between first and second states, the switchable polarizer is configured to transmit, in the first state, the first polarization of the image light through the second surface of the waveguide, and the switchable polarizer is configured to transmit, in the second state, the second polarization of the image light through the second surface of the waveguide.

(91) In accordance with another embodiment, the display module includes a first display panel configured to transmit the first polarization of the image light towards the reflective input coupling prism and the display module includes a second display panel configured to concurrently transmit the second polarization of the image light towards the reflective input coupling prism.

(92) In accordance with another embodiment, the reflective input coupling prism has a reflective surface configured to reflect the image light into the waveguide, the waveguide has opposing first and second surfaces, the reflective input coupling prism includes a first wedge mounted to the first surface and a second wedge mounted to a surface of the first wedge, the second wedge includes the reflective surface, the surface of the first wedge is configured to reflect the image light into the

waveguide in a first direction, and the reflective surface is configured to reflect the image light into the waveguide in a second direction that is different from the first direction, the display includes a switchable reflective layer interposed between the first and second wedges, the switchable reflective layer is switchable between first and second states, the switchable reflective layer is configured to reflect, in the first state, the image light in the first direction, the switchable reflective layer is configured to transmit, in the second state, the image light towards the reflective surface, and the switchable reflective layer includes a cholesteric liquid crystal reflector.

(93) In accordance with another embodiment, the display includes a birefringent beam displacer configured to receive a pupil of image light, the birefringent beam displacer is configured to transmit a first polarization of the pupil towards the reflective input coupling prism within a first beam, and the birefringent beam displacer is configured to transmit a second polarization of the pupil towards the reflective input coupling prism in a second beam that is offset from the first beam.

(94) In accordance with another embodiment, the waveguide has opposing first and second surfaces, the reflective input coupling prism includes a first wedge mounted to the first surface and a second wedge mounted to a surface of the first wedge, the birefringent beam displacer is mounted to the second surface, the first wedge has a first refractive index, and the second wedge has a second refractive index that is different from the first refractive index.

(95) In accordance with another embodiment, the reflective input coupling prism includes a birefringent beam displacer.

(96) In accordance with another embodiment, the display includes a lens configured to direct the image light from the display module towards the reflective input coupling prism, the lens includes a first set of lens elements, and a second set of lens elements separated from the first set of lens elements by a gap, the second set of lens elements is interposed between the first set of lens elements and the waveguide, and the first set of lens elements is configured to converge the image light at a location within the gap.

(97) In accordance with another embodiment, the display includes a switchable element at the location within the gap, the switchable element is switchable between a first state at which the switchable element passes the image light to the second set of lens elements with a first alignment and a second state at which the switchable element passes the image light to the second set of lens elements with a second alignment that is displaced with respect to the first alignment, and the switchable element includes a transparent panel and a piezoelectric component configured to rotate the transparent panel between a first orientation in the first state and a second orientation in the second state.

(98) In accordance with an embodiment, an electronic device is provided that includes a display module that produces image light, a waveguide having opposing first and second surfaces, collimating optics that direct the image light towards the second surface of the waveguide, and an input coupling prism configured to receive the image light from the collimating optics through the waveguide, the input coupling prism includes a first wedge mounted to the first surface of the waveguide and having a first reflective surface oriented at a first angle with respect to the first surface of the waveguide, the first wedge is configured to reflect the image light into the waveguide in a first direction, and a second wedge mounted to the first reflective surface and having a second reflective surface oriented at a second angle with respect to the first surface of the waveguide, the second angle is greater than the first angle, the second wedge is configured to receive the image light through the first wedge, and the second wedge is configured to reflect the image light into the waveguide in a second direction that is different from the first direction.

(99) In accordance with another embodiment, the electronic device includes a polarizing beam splitter layered on the first reflective surface.

(100) In accordance with another embodiment, the electronic device includes a switchable reflective layer on the first reflective surface.

- (101) In accordance with another embodiment, the first wedge has a first refractive index and the second wedge has a second refractive index that is different from the first refractive index.
- (102) In accordance with another embodiment, the second reflective surface is curved and the first reflective surface is curved.
- (103) In accordance with an embodiment, a display is provided that includes a display module that generates image light, a waveguide having opposing first and second surfaces, a lens configured to direct the image light towards the second surface of the waveguide, the lens includes a set of lens elements configured to converge the image light at a location interposed between the set of lens elements and the second surface of the waveguide, and a reflective input coupling prism mounted to the first surface of the waveguide and configured to receive the image light from the lens through the waveguide, the reflective input coupling prism is configured to reflect the image light into the waveguide.
- (104) In accordance with another embodiment, the lens includes an additional set of lens elements interposed between the location and the second surface of the waveguide, including a transparent panel at the location, the transparent panel is switchable between first and second orientations, the transparent panel is configured to pass, in the first orientation, the image light to the second set of lens elements with a first alignment, and the transparent panel is configured to pass, in the second orientation, the image light to the second set of lens elements with a second alignment that is offset from the first alignment.
- (105) The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device comprising: a projector configured to output light; a waveguide; an input coupler comprising a prism configured to couple the light into the waveguide, the waveguide being configured to propagate the light via total internal reflection; and an optical wedge optically coupled between the projector and the waveguide, wherein the optical wedge is configured to direct the light from the projector towards the input coupler.
2. The electronic device of claim 1, wherein the optical wedge has a first surface configured to transmit the light and a second surface opposite the first surface and configured to transmit the light, the second surface being non-parallel with respect to the first surface.
3. The electronic device of claim 2, wherein the first surface faces the waveguide and is parallel to a first lateral surface of the waveguide.
4. The electronic device of claim 3, wherein the waveguide has a second lateral surface opposite the first lateral surface, wherein the prism is mounted to the second lateral surface, and wherein the first lateral surface is parallel to the second lateral surface.
5. The electronic device of claim 1, further comprising: an output coupler configured to couple the light out of the waveguide.
6. The electronic device of claim 1, wherein the optical wedge comprises a dispersion compensation wedge configured to compensate for dispersion of the light by the input coupler.
7. The electronic device of claim 1, wherein the prism is a reflective prism with a reflective surface configured to reflect the light into the waveguide.
8. The electronic device of claim 7, further comprising: a reflective layer that covers a portion of the reflective surface that overlaps a pupil of the light that has been focused onto the reflective surface; and an optical absorber that covers a remainder of the reflective surface.
9. A display comprising: a projector configured to generate light; a waveguide; an input coupler, wherein the input coupler is configured to couple the light into the waveguide; and a birefringent beam displacer configured to receive a pupil of the light from the projector, wherein the

birefringent beam displacer is configured to transmit a first polarization of the pupil towards the input coupler within a first beam, wherein the birefringent beam displacer is configured to transmit a second polarization of the pupil towards the input coupler in a second beam that is offset from the first beam, and wherein the second polarization is different from the first polarization.

10. The display of claim 9, wherein the birefringent beam displacer is optically interposed between the waveguide and the projector.

11. The display of claim 10, wherein the birefringent beam displacer is mounted to a surface of the waveguide.

12. The display of claim 9, wherein the waveguide is interposed between the birefringent beam displacer and the input coupler.

13. The display of claim 9, wherein the birefringent beam displacer is configured to split the pupil of the light from the projector into the first beam and the second beam.

14. The display of claim 9, wherein the first beam and the second beam collectively exhibit an expanded pupil of light that is larger than the pupil of light.

15. A display comprising: a projector that produces light; a waveguide having opposing first and second surfaces, wherein the waveguide is configured to propagate the light via total internal reflection; and an input coupling prism configured to receive the light, wherein the input coupling prism comprises: a first wedge mounted to the first surface of the waveguide, wherein the first wedge is configured to couple the light into the waveguide in a first direction, and a second wedge mounted to a surface of the first wedge, wherein the second wedge is configured to couple the light into the first wedge in a second direction that is non-parallel to the first direction.

16. The display of claim 15, further comprising: a polarizing beam splitter or a switchable reflective layer interposed between the first and second wedges.

17. The display of claim 15, further comprising collimating optics that direct the light towards the second surface of the waveguide, wherein the input coupling prism is configured to receive the light from the collimating optics.

18. The display of claim 15, wherein the surface of the first wedge is oriented at a first angle with respect to the first surface of the waveguide, the second wedge has an additional surface oriented at a second angle with respect to the first surface of the waveguide, and the second angle is greater than the first angle.

19. The display of claim 15, wherein the surface of the first wedge is curved and wherein an additional surface of the second wedge is curved.

20. A display comprising: a projector that produces light; a waveguide having opposing first and second surfaces; and an input coupling prism configured to receive the light, wherein the input coupling prism comprises: a first wedge with a first refractive index and mounted to the first surface of the waveguide, wherein the first wedge is configured to couple the light into the waveguide in a first direction, and a second wedge with a second refractive index different from the first refractive index and mounted to a surface of the first wedge, wherein the second wedge is configured to couple the light into the first wedge in a second direction that is non-parallel to the first direction.
